

# PHYSICS PRACTICAL SHEETS

.....KV.....-Campus

Date 18 June 2026

Class : CE-1/11

Roll No.: 43

Shift: Morning

Object of the Experiment (Block Letter)

*Caustic*

Experiment No.: 3

Group : 5

Sub: PHYS 102 lab

Set: 1

## DETERMINATION OF SURFACE TENSION OF WATER BY CAPILLARY RISE METHOD

Apparatus required:

- 1) Capillary tubes of uniform bore & different radii
- 2) A glass strip
- 3) A thin rubber band
- 4) A needle
- 5) A glass vessel
- 6) A clamp stand
- 7) Travelling microscope
- 8) An adjustable stand

Theory:

When a capillary tube is dipped in water, the liquid rises within the tube until the downward gravitational force of the liquid column is balanced by the upward vertical component of the surface tension. At equilibrium, the weight of the fluid column is equal to the vertical component of the surface tension demonstrating a balance of forces, so

$$2r\gamma \cos \theta = \pi r^2 \left( \frac{h + r}{3} \right) \rho g \quad \text{--- (1)}$$

where  $r$  is the radius of the tube,  $h$  is the height of the liquid column,  $\rho$  is the density of water column and  $\gamma$  is the surface tension of water.



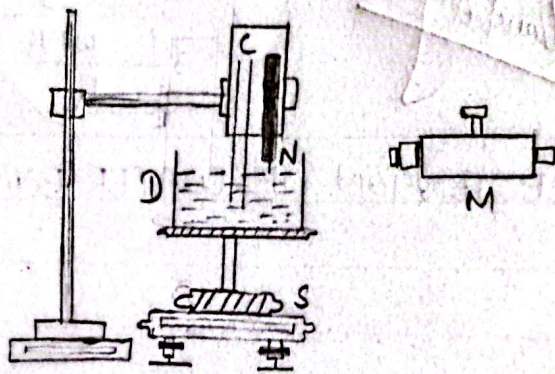


Fig 1: Schematic diagram of the experimental setup used to measure surface tension of a liquid (M: Microscope, N: Needle, C: capillary tube, A: Beaker, S: Stand)

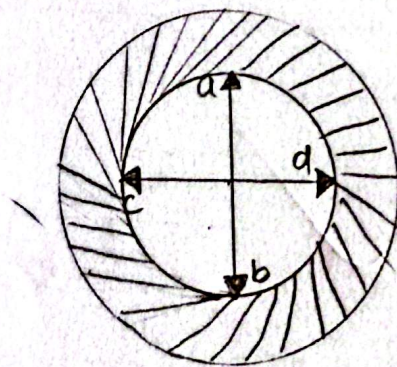


Fig 2: Cross sectional view of a capillary tube



The relation can be simplified by considering the radius of the capillary to be small in comparison to the height of the water column & the angle of contact  $\theta$  is small (8-9 degrees for water glass surface).

The surface tension of the liquid is determined by the formula.

$$\gamma = \frac{r h \rho g}{2} \quad \text{--- (2)}$$

### OBSERVATIONS

Temperature of water,  $T = 25^\circ\text{C}$

Least count of main scale,  $M = 1/20 \text{ cm} = 0.05 \text{ cm}$

Number of Vernier division,  $N = 50$

Vernier constant  $\frac{M}{N} = \frac{1}{20 \times 50} = 0.001 \text{ cm}$

Density of water at  $T = 1 \text{ gm/cm}^3$

Table 1: Height of water column in capillary rise method

| Tube No. | No. of obs | Position of the meniscus |               |                |           | Position of the needle tip |               |                |           | Height $h = X - Y$ (cm) |
|----------|------------|--------------------------|---------------|----------------|-----------|----------------------------|---------------|----------------|-----------|-------------------------|
|          |            | Values of Main scale     | Vernier scale | Total $X$ (cm) | Mean (cm) | Values of Main scale       | Vernier scale | Total $Y$ (cm) | Mean (cm) |                         |
| I        | 1          | 5.55                     | 0.014         | 5.564          | 5.532     | 3.35                       | 0.011         | 3.36           | 3.4175    | 2.1145                  |
|          | 2          | 5.45                     | 0.05          | 5.5            |           | 3.45                       | 0.025         | 3.475          |           |                         |
| II       | 1          | 5.35                     | 0.02          | 5.37           | 5.397     | 3.4                        | 0.04          | 3.44           | 3.331     | 2.0665                  |
|          | 2          | 5.4                      | 0.025         | 5.425          |           | 3.2                        | 0.022         | 3.222          |           |                         |



Table 2 : Diameter of the capillary tube:

| No. of<br>obs | Tube<br>No. | Microscopic reading (cm) |      |      |       | Diameter (cm) |       | Mean<br>Diameter (cm) | Mean<br>radius<br>r (cm) |
|---------------|-------------|--------------------------|------|------|-------|---------------|-------|-----------------------|--------------------------|
|               |             | a                        | b    | c    | d     | 1b-a1         | 1d-c1 |                       |                          |
| 1             | I           | 6.04                     | 5.99 | 7.98 | 7.846 | 0.126         | 0.13  | 0.128                 | 0.064                    |
| 2             | II          | 6.087                    | 5.94 | 7.87 | 7.722 | 0.943         | 0.143 | 0.144                 | 0.072                    |

**RESULT:**

The value of the surface tension will be :

For tube I :  $\gamma^2 = 66.31 \text{ dyne/cm}$

For tube II :  $\gamma^2 = 72.91 \text{ dyne/cm}$

The mean value of surface tension of water is  $69.61 \text{ dyne/cm}$ .

The surface tension of water (at  $T=25^\circ\text{C}$ ) is  $72 \text{ dyne/cm}$ .

**Precautions :**

- ✓ Use only distilled water
- ✓ Use a clean container for the water
- ✓ Precisely measure the capillary tube's radius.
- ✓ Use meticulously clean capillary tubes with a uniform bore.
- ✓ Maintain and record a constant water temperature.
- ✓ Ensure the capillary tube is perfectly vertical.

**Conclusion :**

The value of surface tension of water was determined from the experiment.

*Darshini*  
25th June 2026.